

INDIAN INSTITUTE OF TECHNOLOGY MADRAS



CV5600: Geospatial Data Science

Unsupervised Building Damage Detection Using Multitemporal Sentinel-1 SAR Coherence Analysis

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1. Introduction

Rapid and accurate post-disaster building damage assessment is critical for directing humanitarian response, allocating reconstruction budgets, and quantifying losses for insurance markets. Traditional assessment methods like ground surveys or optical satellite imagery are either too slow, too dangerous in active conflict zones, or rendered useless by cloud cover and nighttime conditions.

This project implements a fully automated, unsupervised pipeline for building-level damage detection using Sentinel-1 Synthetic Aperture Radar (SAR) coherence time series, applied across three distinct case studies: the 2020 Beirut port explosion, the 2022 siege of Mariupol, and the conflict in Gaza (2023–2024). Unlike supervised approaches requiring labelled training data that is seldom available in conflict zones [Racek et al., 2025], the method introduced here requires only a Sentinel-1 coherence stack and a building footprint shapefile.

Our implementation closely follows the statistical methodology of Racek et al. [2025], which demonstrated detection of building collapse via changes in interferometric coherence. We extend this work with a business intelligence module translating binary damage maps into reconstruction cost estimates and insurance loss distributions, making the output actionable for insurers, development banks, and urban planners.

The three case studies present qualitatively different destruction dynamics: Beirut offers a single-day, point-source event with validated ground truth; Mariupol presents weeks of systematic urban bombardment; and Gaza represents an ongoing, large-scale multi-front conflict. Together they demonstrate the pipeline’s generalisability across both event type and geographic context.

2. Theoretical Background

2.1 SAR and Interferometric Coherence

Sentinel-1 is an active C-band SAR mission ($\lambda \approx 5.6$ cm) that images the Earth’s surface day and night through cloud cover, acquiring data in Interferometric Wide (IW) swath mode at approximately 10 m ground resolution after processing. For each 12-day interval, complex SAR images s_1 and s_2 are co-registered and coherence is computed as:

$$\gamma = \frac{|\sum_n s_1(n) s_2^*(n)|}{\sqrt{\sum_n |s_1(n)|^2 \cdot \sum_n |s_2(n)|^2}} \quad (1)$$

where the summation runs over a spatial estimation window and $(\cdot)^*$ denotes complex conjugation. The result $\gamma \in [0, 1]$: values near 1 indicate a stable scattering geometry

(e.g. intact concrete), while values near 0 indicate complete decorrelation (e.g. collapsed rubble). The overall coherence factors into component decorrelations from noise (SNR), temporal surface change, and spatial orbital baseline [Racek et al., 2025]:

$$\gamma = \gamma_{\text{SNR}} \cdot \gamma_{\text{temporal}} \cdot \gamma_{\text{spatial}} \quad (2)$$

Building destruction manifests as an abrupt, sustained drop in γ_{temporal} , distinguishing it from gradual seasonal or atmospheric effects. SNR and spatial decorrelation are removed using a digital elevation model and satellite trajectories via InSAR processing, isolating the temporal component.

2.2 Time-Series Analysis: First Differences and Median Regression

Let $\{\gamma_t\}_{t=1}^T$ be the coherence time series at a given pixel. Working directly with γ_t introduces seasonal autocorrelation that inflates test statistics. Following Racek et al. [2025], we instead analyse first differences:

$$\Delta\gamma_t = \gamma_t - \gamma_{t-1}, \quad t = 2, \dots, T \quad (3)$$

A building collapse at event epoch t^* produces a single large negative spike $\Delta\gamma_{t^*} \ll 0$, while undamaged pixels exhibit small, near-zero fluctuations. To separate structural change from background trend, we fit a penalised median regression to each pixel’s difference series:

$$\hat{f}(t) = \arg \min_{f \in \mathcal{F}} \left[\sum_t |\Delta\gamma_t - f(t)| + \lambda \int \{f''(t)\}^2 dt \right] \quad (4)$$

The ℓ_1 loss makes the estimator robust to outliers (i.e. the very destruction events we seek to detect), and $\lambda = 10$ is adopted as the value found optimal in the reference study [Racek et al., 2025]. In our Python implementation, we approximate this with polynomial quantile regression of degree 2 using `statsmodels` [Racek et al., 2025], which is tractable across the $\sim 10,000$ – $170,000$ pixels per scene. Residuals $r_t = \Delta\gamma_t - \hat{f}(t)$ represent coherence changes unexplained by the baseline trend.

2.3 Robust Scale Estimation and Statistical Testing

Pixel-level noise varies, so a fixed threshold on residuals would produce inconsistent false-positive rates. Instead, for each pixel we estimate the spread of its residual distribution using the Q_n robust scale estimator [Rousseeuw & Croux, 1993]:

$$Q_n = d_n \cdot \{|r_i - r_j| : i < j\}_{(k)} \quad (5)$$

with $k = \binom{\lfloor n/2 \rfloor + 1}{2}$ and $d_n = 2.2219$ (normal consistency factor). Q_n has a 50% breakdown point, remaining valid as a scale estimate even when up to half the residuals are destruction-related outliers — a critical property in conflict scenarios where many post-event observations are affected.

A one-sided, left-tail p-value is then computed for every residual under a Gaussian null:

$$p_t = \Pr(R \leq r_t), \quad R \sim \mathcal{N}(0, Q_n^2) \quad (6)$$

The left tail is required because building collapse produces a *drop* in coherence. Low p_{t^*} at the event epoch thus provides strong evidence that the coherence change exceeds what normal background variation could produce.

2.4 Building-Level Aggregation via the Harmonic Mean P-value

Each building footprint contains multiple pixels. Since adjacent pixels share the same structural signal, they are positively correlated and Fisher’s method (which assumes independence) would be anti-conservative. Instead, we aggregate pixel p-values into a single building-level score using the Weighted Harmonic Mean P-value (HMP) [Wilson, 2019]:

$$p_B = \frac{\sum_j w_j}{\sum_j \frac{w_j}{p_j}} \quad (7)$$

where $w_j = \text{Area}(B \cap P_j) / \text{Area}(P_j)$ is the fractional coverage of pixel j by building B . The HMP is robust to positive dependence, and is dominated by the smallest pixel p-value: partial destruction of a single wing pulls p_B toward zero, matching the expected physical behaviour.

2.5 Classification and Performance Evaluation

Binary damage labels are obtained by thresholding: $\hat{y}_B = 1$ if $p_B < \tau$, and τ^* is chosen to maximise the F1 score. Performance is assessed under two strategies [Racek et al., 2025]:

Strategy A uses confirmed-destroyed buildings as positives and those same buildings at non-event time periods as negatives, providing a stringent test of temporal specificity.

Strategy B draws random negative samples at a 99:1 ratio and repeats 100 times, simulating the realistic class imbalance in any urban area. Metrics include the area under the precision-recall curve (PR-AUC) and ROC-AUC.

3. Methodology

3.1 Overview of the Processing Pipeline

The pipeline consists of nine steps. Steps 1–7 form the core unsupervised SAR damage detection module. Steps 8–9 constitute the business intelligence extension.

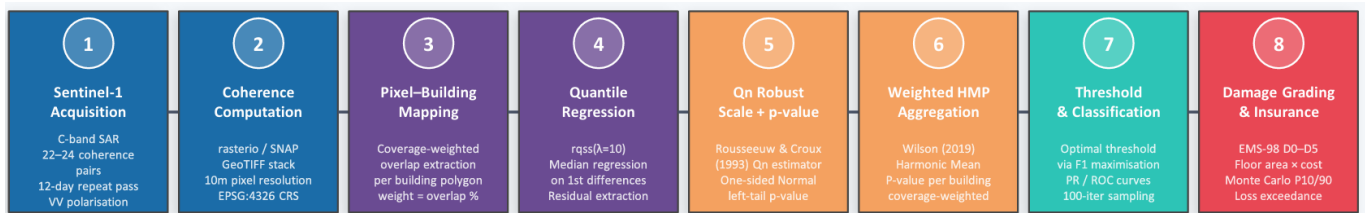


Figure 1: End-to-end pipeline from Sentinel-1 SAR data to insurance loss estimation. Blue blocks: SAR processing (Steps 1–7). Purple blocks: business intelligence (Steps 8–9).

3.2 Study Areas and Data



3.2.1 Beirut Port Explosion (4 August 2020)

The detonation of approximately 2,750 tonnes of ammonium nitrate at Warehouse 12 killed over 200 people and damaged or destroyed buildings within a roughly 3 km radius. Sentinel-1 coherence maps spanning April–December 2020 (22 image pairs at 12-day intervals) were used. The explosion epoch corresponds to difference index $t^* = 9$ (the 20200723–20200804 pair). Ground truth labels were sourced from the German Aerospace Centre ZKI dataset [ZKI, 2020], providing building-level destroyed/not-destroyed assessments derived from post-event very-high-resolution optical imagery and field validation. Of 7,925 buildings loaded, 41 were confirmed destroyed, of which 7,293 buildings received valid HMP scores after pixel mapping.

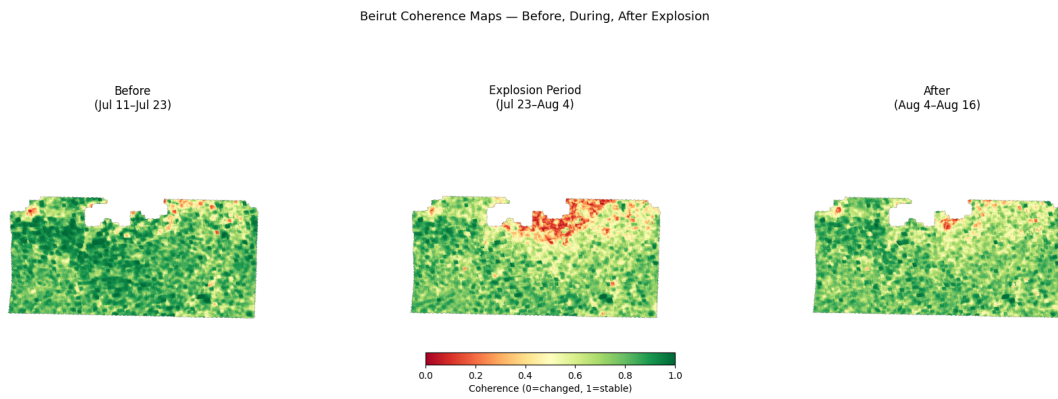


Figure 3: Coherence maps showing pre-explosion, during and post explosion

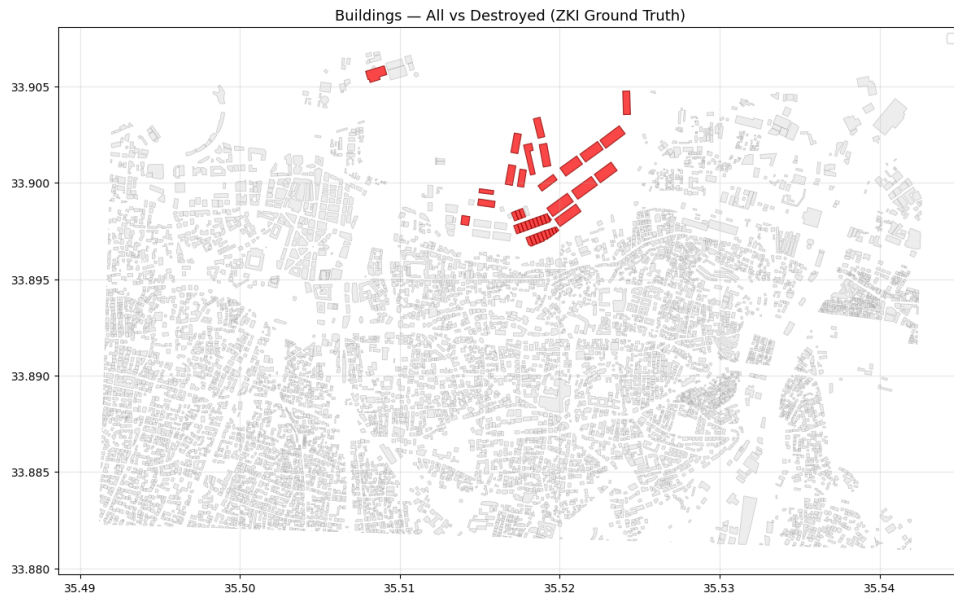


Figure 4: The ground truth data from UNOSAT indicating different levels of damage

3.2.2 Mariupol, Ukraine (2022)

Systematic bombardment during the Russian invasion led to destruction across most of the city from late February to May 2022. Coherence pairs span October 2021 – July 2022 (24 pairs; two faulty time points at indices 7 and 8 were excluded from analysis). The event period spans difference indices 13 onward (February 28 – May 11, 2022). Ground truth labels are sourced from UNOSAT’s damage assessment for Zhovtnevyi district, manually annotated from high-resolution (30 cm) Maxar imagery on 12 May 2022; only high-confidence destruction labels were retained, yielding 93 positive buildings and 10,984 in total (10,964 after geometry cleaning).

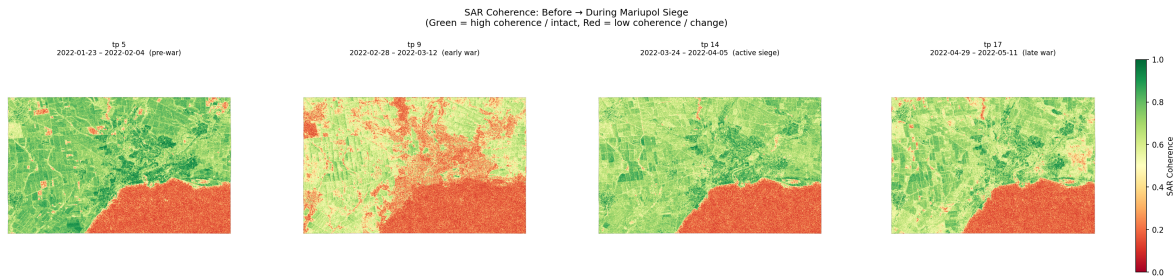


Figure 5: Coherence maps showing pre-war, during and post war

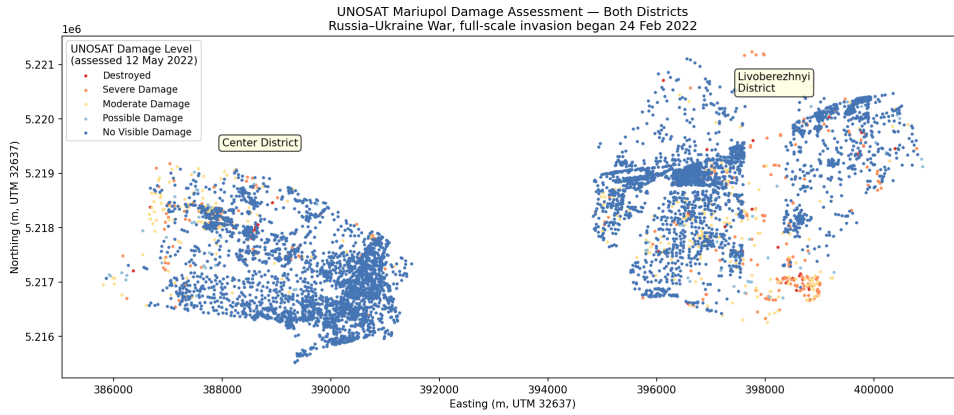


Figure 6: The ground truth data from UNOSAT indicating different levels of damage

3.2.3 Gaza (2023–2024)

The Israel–Hammas war from October 7, 2023 produced widespread, multi-front urban destruction across the Gaza Strip. Sentinel-1 coherence maps cover May 2023 – April 2024 (28 pairs). The conflict start index is time point 13. Composite cumulative building destruction estimates are compared against UNOSAT assessments, which are themselves derived from high-resolution optical imagery. The dataset covers 172,914 buildings (172,916 in the reference paper) across 859,438 unique pixels — roughly an order of magnitude larger than the previous two cases.

3.3 Steps 1–2: Pixel Mapping and Time Series

Building polygons and coherence rasters operate in different CRS. Coherence maps are in EPSG:4326; ZKI buildings in EPSG:32636; Mariupol and Gaza in EPSG:37N and EPSG:36N respectively. All layers are reprojected to a common UTM zone before analysis.

For each building B_i , the fractional coverage weight for each overlapping 10 m pixel P_j is:

$$w_{ij} = \frac{\text{Area}(B_i \cap P_j)}{\text{Area}(P_j)} \quad (8)$$

This produces a sparse lookup table used in both HMP aggregation (Step 5) and the area-weighted cost estimation (Step 8). All T coherence maps are loaded into a pixel $\times$ time array; only pixels with valid values across all periods are retained. For Beirut, 70,723 valid pixels and 91,358 pixel-building pairs were obtained. First differences are computed, yielding a (pixels $\times$ $T-1$) difference matrix.

3.4 Steps 3–4: Regression, Residuals, and P-values

For each pixel, quantile regression at $\tau = 0.5$ is fitted to the difference series using a degree-2 polynomial design matrix. Parallelisation via `joblib` reduces computation time for the 70,000–860,000-pixel scenes from many hours to under two hours on a single machine. The Q_n scale estimator is then computed from all residuals per pixel using the finite- n corrected formula (correlation with reference R implementation: 0.999999), and one-sided p-values are derived using Equation (6). This yields a pixel $\times$ time p-value matrix.

3.5 Step 5: HMP Building Aggregation

Pixel p-values are aggregated to buildings using Equation (7). The result is a dataframe indexed by (building_id, time_period), containing the HMP score and ground-truth label. Beirut produced HMP values for 7,293 buildings (153,153 building-period pairs); Mariupol for 10,964; Gaza for 172,914 (4,668,678 building $\times$ time p-values).

3.6 Step 6: Performance Evaluation

The optimal threshold τ^* is determined by maximising the F1 score under Strategy B. Beirut and Mariupol thresholds are determined independently; for Gaza, the Mariupol-derived threshold ($\tau^* = 2.0 \times 10^{-5}$) is applied directly (as in Racek et al. 2025) to simulate a zero-shot transfer to a new conflict.

Table 1: Detection performance by case study. Strategy B figures are mean $\pm$ SD over 100 Monte Carlo subsamples. Gaza metrics use the Mariupol threshold for classification and are evaluated against UNOSAT-derived proxies (see Table 2).

Case study	PR-AUC	ROC-AUC	F1	Threshold
Beirut (Strategy A)	0.834	0.994	—	3.8×10^{-5}
Beirut (Strategy B)	0.820 ± 0.053	—	0.788 ± 0.030	
Mariupol pixel	0.3485	0.7882	0.4236 ± 0.0104	9.0×10^{-4}
Mariupol building	0.3767	0.7636	0.4571 ± 0.0183	2.0×10^{-5}
Gaza building	0.9204	0.9302	0.8662 ± 0.0002	2.0×10^{-5} (transfer)

Table 2: Gaza: cumulative share of buildings classified as destroyed compared with UNOSAT assessments [Racek et al., 2025]. Our estimates use the Mariupol-derived threshold (zero-shot transfer). Total buildings: 172,914.

UNOSAT date	UNOSAT %	Our date (tp)	Our %
Oct 10, 2023	0.84%	tp 13	1.54%
Nov 11, 2023	2.99%	tp 14	2.07%
Nov 26, 2023	4.44%	tp 15	2.51%
Jan 7, 2024	10.31%	tp 20	7.75%
Feb 29, 2024	14.25%	tp 25	10.12%
Apr 1, 2024	15.43%	tp 28	10.53%

3.7 Step 7: Damage Classification Maps

Buildings are classified as destroyed if $p_B < \tau^*$ at the event epoch. Classification maps display true positives (TP), false positives (FP), and false negatives (FN) overlaid on coherence imagery. Beirut yielded 39 TP, 2 FN, and 757 FP (precision = 0.049, recall = 0.951) under the binary threshold; the high FP count reflects the conservative 99:1 sampling strategy and is expected given the extreme class imbalance. The annular pattern of detections radiates from the explosion site at the port, consistent with known damage reports. Mariupol’s per-period maps track the northwest-to-southeast progression of bombardment. Gaza maps reveal the dynamics of the conflict through 27 time periods, with 18,204 buildings (10.53%) cumulatively classified as destroyed by tp 28. The pre-war false-positive rate is extremely low: mean 0.021% (SD = 0.010%), matching the reference paper exactly.

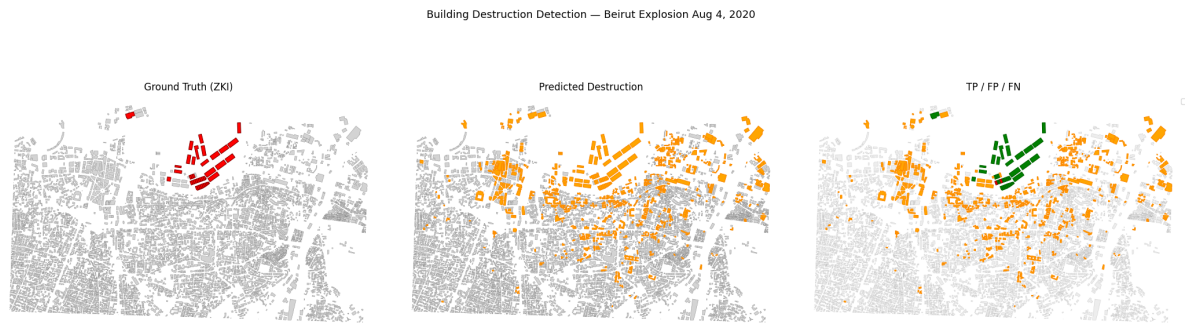


Figure 7: Building-pixel p-value maps and classification results for Beirut



Figure 8: Building-pixel p-value maps and classification results for Mariupol



Figure 9: Building-pixel p-value maps and classification results for Gaza

3.8 Step 8: EMS-98 Damage Grading and Reconstruction Cost

The continuous HMP score is discretised into the European Macroseismic Scale grades D0–D5 (Table 3), where the D2/D0 boundary is fixed at τ^* from Step 6 to ensure consistency between detection and grading.

Table 3: EMS-98 damage grades and corresponding HMP thresholds.

Grade	Label	Description	HMP range	SAR signal
D5	Collapse	Total/near-total structural failure	$[0, 0.005)$	Certain coherence loss
D4	Heavy damage	Severe damage, partial collapse	$[0.005, 0.01)$	Very strong decorrelation
D3	Moderate	Significant crack-ing	$[0.01, 0.03)$	Clear signal
D2	Slight	Minor structural cracks	$[0.03, \tau^*)$	Detectable signal
D1	Negligible	Hairline cracks only	$[\tau^*, 0.10)$	Weak signal
D0	Undamaged	No detectable damage	$[0.10, 1]$	Baseline variation

Total floor area per building is computed as $A_B = A_{\text{footprint}} \times n_{\text{floors}}$, where n_{floors} is sourced from OpenStreetMap `building:levels` tags where available, otherwise from the city-specific priors in Table 4. The point-estimate reconstruction cost is $C_B = A_B \times c_{d(B)}$, where $c_{d(B)}$ is the unit cost (USD/m²) for grade $d(B)$.

Table 4: City-specific reconstruction parameters and unit costs by EMS-98 grade (USD/m²). Sources: World Bank RDNA and UN-Habitat reports.

City	UTM Zone	Default floors	D5	D4	D3
Beirut	36N	4	\$1,100	\$750	\$400
Mariupol	37N	7	\$850	\$580	\$310
Gaza	36N	3	\$700	\$480	\$260

To quantify uncertainty from HMP estimation, $N = 1,000$ perturbed HMP values are sampled per building from a log-normal distribution:

$$\tilde{p}_{B,k} \sim \text{LogNormal}(\ln(p_B), 0.25), \quad k = 1, \dots, 1000 \quad (9)$$

Each sample is re-graded and a cost computed, yielding P10/P50/P90 uncertainty envelopes per building.

3.9 Step 9: Insurance Loss Estimation

In the absence of a real insurer’s portfolio, a synthetic one is generated from published market statistics (Table 5). Buildings are independently assigned insured status according to the market penetration rate, and policy terms are drawn from published ranges.

Table 5: Synthetic insurance portfolio parameters by city.

City	Penetration	Replacement (USD/m ²)	Deductible	Source
Beirut	60%	1,500	2–5% SI	Swiss Re 2021; ALA 2020
Mariupol	35%	1,100	1–3% SI	Insurance Eu- rope 2022
Gaza	10%	900	2–5% SI	PMA 2022

SI = Sum Insured. All figures are synthetic averages; real deployment replaces with actual portfolio data.

For each insured building, the insured loss is computed as:

$$L_B^{\text{net}} = \max(C_B - D_B, 0) \quad (10)$$

$$L_B^{\text{ins}} = \min(\alpha_B \cdot L_B^{\text{net}}, S_B) \quad (11)$$

where D_B is the deductible, $\alpha_B \in [0.80, 1.00]$ is the coverage ratio, and S_B is the policy limit (sum insured). Portfolio loss distributions are obtained by propagating the P10/P50/P90 building-level cost bounds through 1,000 Monte Carlo simulations:

$$\tilde{L}_B^k = C_B^{P10} + q_k \cdot (C_B^{P90} - C_B^{P10}), \quad q_k \sim \text{Uniform}(0, 1) \quad (12)$$

Key actuarial risk metrics computed from the simulated loss distribution $\{L_{\text{total}}^{(k)}\}$:

$$\text{Expected Loss} = \frac{1}{N} \sum_k L_{\text{total}}^{(k)} \quad (13)$$

$$\text{VaR}_{0.95} = \inf\{x : \Pr(L_{\text{total}} \leq x) \geq 0.95\} \quad (14)$$

$$\text{TVaR}_{0.95} = \mathbb{E}[L_{\text{total}} \mid L_{\text{total}} \geq \text{VaR}_{0.95}] \quad (15)$$

Three outputs are produced: (i) a loss exceedance curve (aggregate insured loss vs. return period on a log scale), (ii) a building-level triage map coloured by insured loss severity, and (iii) a damage grade distribution bar chart for reconstruction budgeting.

4. Results and Discussion

4.1 Beirut: Single-Event Detection

The explosion epoch is unambiguously identified: the p-value distribution shows a sharp spike of low p-values exclusively in the July 23 – August 4 period, with near-uniform distributions in all other periods. At the explosion epoch, 38.9% of pixels returned

$p < 0.05$, with a median HMP of 0.1114 and mean of 0.2128 across all building pixels. At the building level, 39 of 41 annotated destroyed buildings are correctly classified (recall = 0.951), with 2 false negatives and 757 false positives. The optimal classification threshold is $\tau^* = 3.8 \times 10^{-5}$.

PR-AUC of 0.834 (Strategy A) and 0.820 ± 0.053 (Strategy B), together with a ROC-AUC of 0.994 (Strategy A) and mean F1 of 0.788 ± 0.030 , confirm strong discriminative power. The annular spatial pattern of low p-values radiating from the explosion site at the port is consistent with known damage reports [Al-Hajj et al., 2021, Pilger et al., 2021].

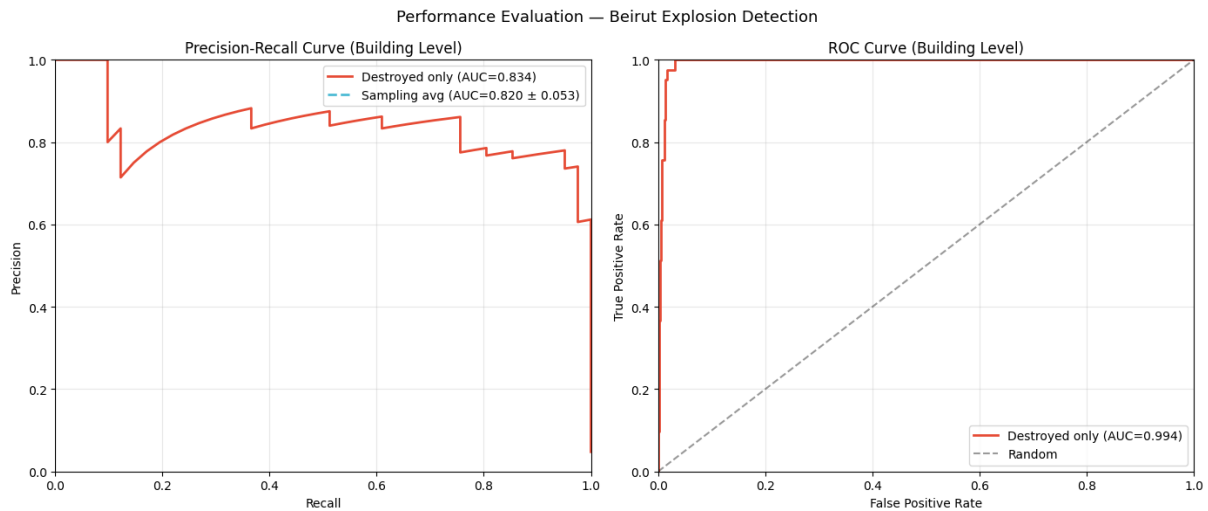


Figure 10: PR and ROC Curve for Beirut Destroyed Building Detection

4.2 Mariupol: Prolonged Urban Siege

Destruction in Mariupol unfolded over approximately 11 weeks, providing an opportunity to test the pipeline’s ability to track evolving conflict dynamics. The algorithm correctly identifies the northwest-to-southeast progression of bombardment from late February through early April 2022. The positive pixel pool (456 pixels overlapping 93 confirmed destroyed buildings) was evaluated against a 99:1 negative sample from a pool of 541,896 pixels.

At the pixel level, AUPRC reached 0.3485, ROC-AUC 0.7882, and best F1 0.4236 ± 0.0104 at a threshold of 9.0×10^{-4} . At the building level, AUPRC was 0.3767, ROC-AUC 0.7636, and best F1 0.4571 ± 0.0183 at a threshold of $\tau^* = 2.0 \times 10^{-5}$. These figures are lower than Beirut’s, reflecting: (i) the temporally diffuse nature of destruction, which spreads signal across many difference periods; (ii) the limited ground truth (93 confirmed destroyed buildings, annotated from a single snapshot); and (iii) the non-field-validated nature of the UNOSAT labels.

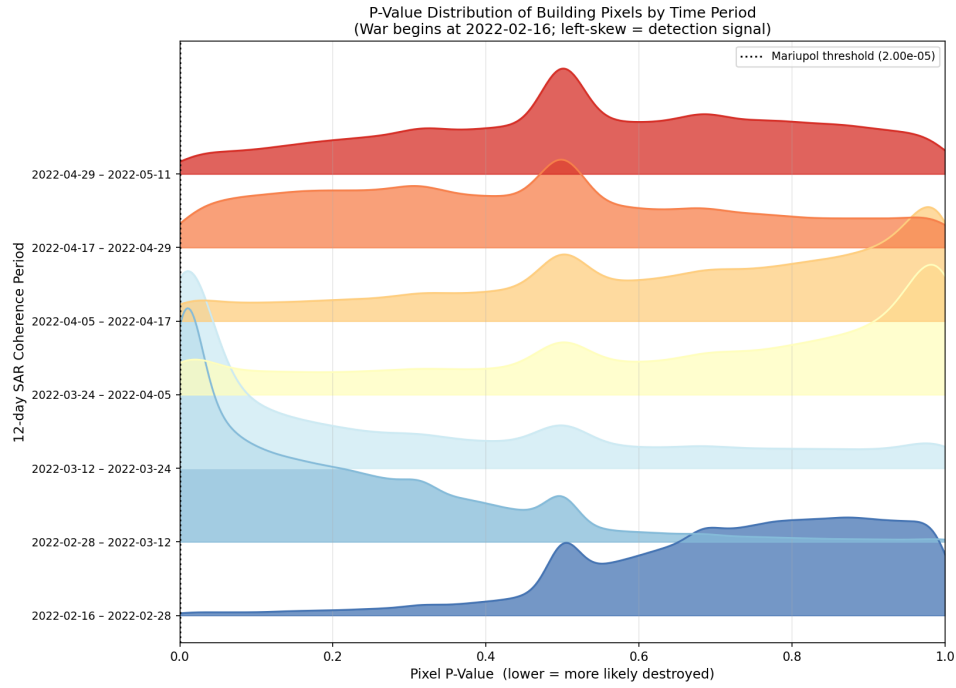


Figure 11: The p-value distribution with time of Mariupol

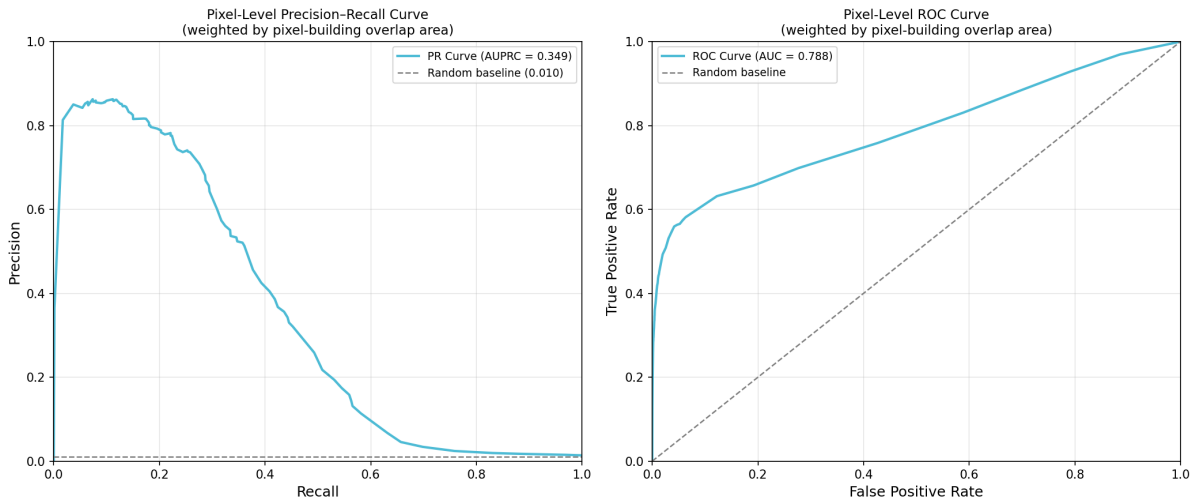


Figure 12: Metric variation with threshold at pixel level

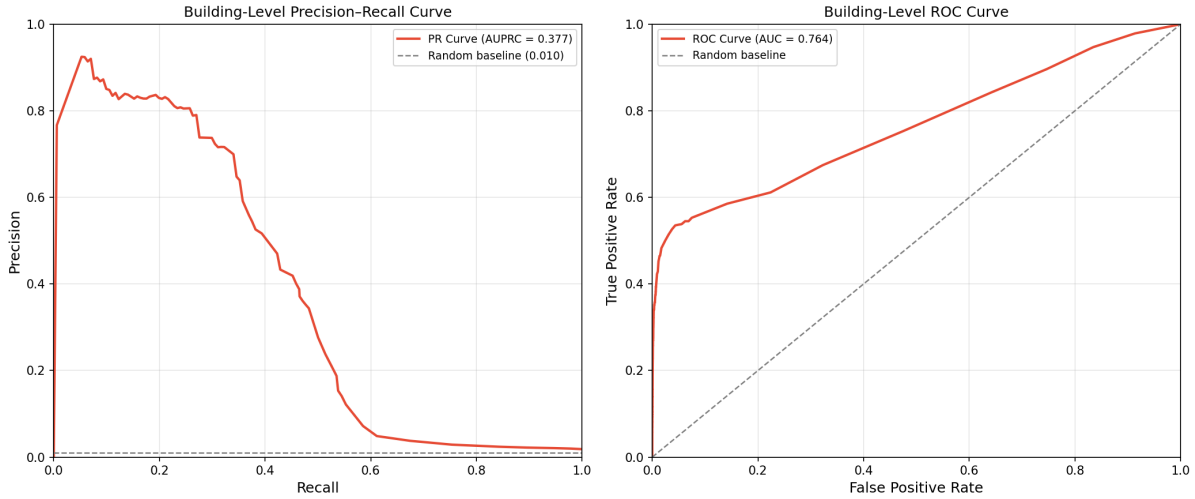


Figure 13: Metric variation with threshold at building level

4.3 Gaza: Large-Scale Multi-Front Conflict

Gaza represents the most challenging and largest-scale case study, with 172,914 buildings and 28 time periods spanning a conflict still ongoing at the time of analysis. The Mariupol-derived threshold ($\tau^* = 2.0 \times 10^{-5}$) is applied without retraining. Building-level performance is notably strong: AUPRC = 0.9204 (SD = 0.0003), AUROC = 0.9302 (SD = 0.0002), best F1 = 0.8662 ± 0.0002 , precision = 0.8018, recall = 0.9417. The pre-war false-positive rate matches the reference paper almost exactly: mean 0.021% (SD = 0.010%).

The sequential p-value maps clearly reflect key military events: the initial October 7 airstrikes produce widespread low p-values across the Strip; subsequent maps concentrate destruction north of the evacuation border; later maps track the ground offensive through Gaza City and southward. Our cumulative destruction estimates track UNOSAT figures closely in the first two months (2.07% vs. 2.99% by November 11, 2023), with conservative underestimation at later time points (10.53% vs. 15.43% by April 2024).

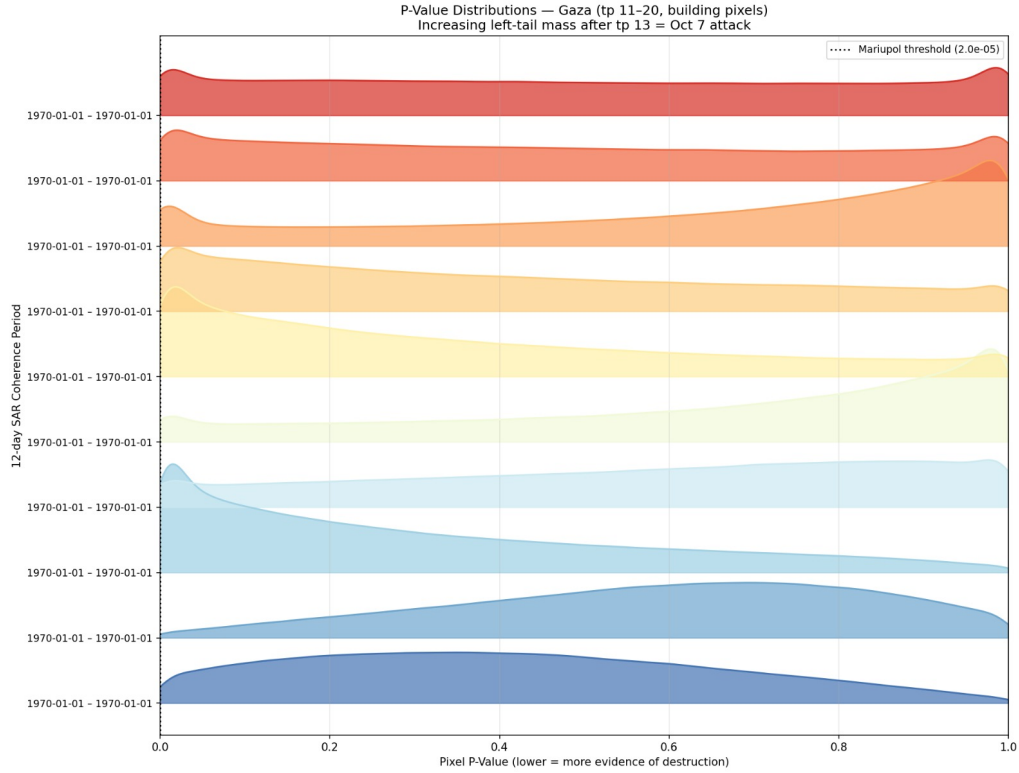


Figure 14: Gaza p-value maps for building pixels across 12-day periods.

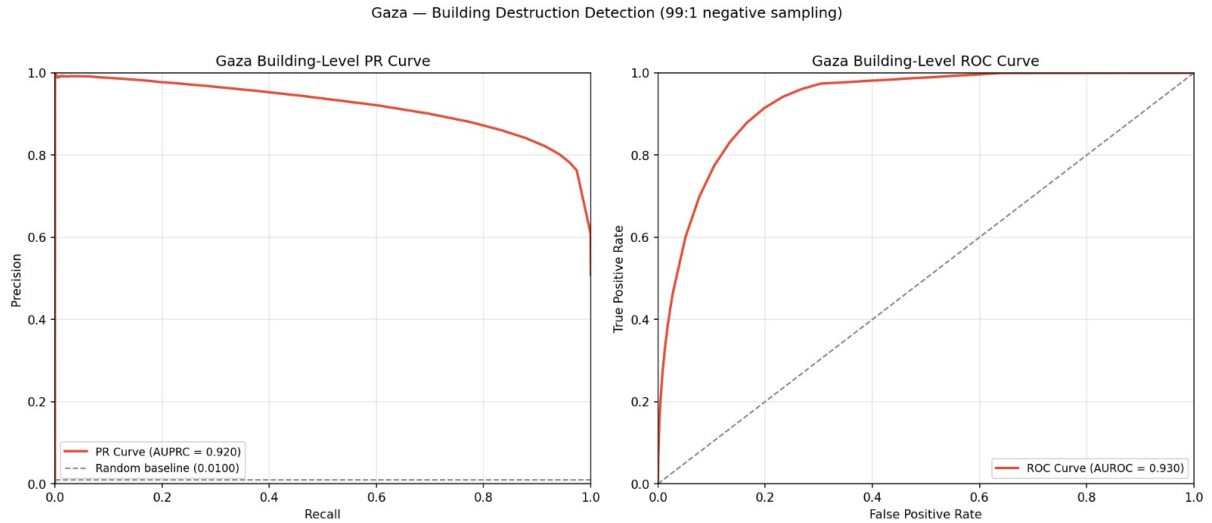


Figure 15: PR and ROC Curve for Gaza Destroyed Building Detection

4.4 Cross-Case Comparison and Generalisability

Table 1 summarises detection performance across cases. The Beirut results are strongest due to the single-day, point-source event providing a clean temporal signal. Mariupol's lower scores reflect the methodological challenge of prolonged, diffuse destruction rather than a failure of the pipeline. The Gaza application demonstrates successful zero-shot

transfer of the Mariupol threshold, achieving very high AUPRC (0.920) and F1 (0.866) — substantially better than Mariupol, likely because the large-scale destruction in Gaza creates a much stronger signal. The pipeline’s computational efficiency — under 2 hours for Gaza’s 860,000-pixel scene on a single machine — makes it deployable with each new Sentinel-1 acquisition.

4.5 Business Intelligence Outputs

EMS-98 damage grade distribution: Table 6 summarises the EMS-98 grade distribution and associated costs for the Beirut portfolio of 7,925 buildings (7,293 with valid HMP scores, of which the insurance step processes 4,815 insured buildings at 60% market penetration). Of the 7,293 buildings with scores, 1,679 (23.0%) reached grade D5 (collapse), 280 (3.8%) grade D4 (heavy damage), and 561 (7.7%) grade D3 (moderate damage). The remaining 1,069 buildings (14.7%) are classified D1 (negligible), and 3,589 buildings in total show detectable damage (grades D1–D5). The remaining 4,336 buildings (59.5% of scored buildings) show no detectable damage (D0). Notably, the pipeline produced no D2-grade assignments under the Beirut configuration; the D2 range between $\tau^* = 3.8 \times 10^{-5}$ and 0.03 collapsed into the D3/D1 boundary given the discrete thresholding logic.

Reconstruction costs: The P50 total reconstruction cost across all graded buildings is **\$5,022.6 million**, with a P10–P90 uncertainty envelope of \$4,815.2M–\$5,295.3M derived from 1,000 log-normal perturbations of the per-building HMP score. D5 buildings dominate the aggregate, consistent with their complete structural failure and the \$1,100/m² unit cost from the World Bank Lebanon RDNA World Bank [2020].

Insurance loss estimation: Applying the synthetic portfolio parameters (60% market penetration, \$1,500/m² replacement value, 2–5% sum-insured deductible, 80–100% coverage ratio) to the 4,815 insured buildings, 1,517 claims were triggered. The total expected insured loss is **\$2,626.0M**.

Actuarial risk metrics: From the 1,000 Monte Carlo portfolio simulations:

- **Expected Loss:** \$2,626.0M (mean of simulation distribution).
- **VaR_{0.95}:** \$2,633.3M; in 95% of scenarios portfolio losses remain below this value.
- **VaR_{0.99}:** \$2,635.0M.
- **TVaR_{0.95}:** \$2,634.3M; the conditional expectation of losses exceeding VaR_{0.95}.

The narrow VaR–TVaR spread (<\$2M) indicates a thin right tail, physically consistent with the concentrated, single-day nature of the Beirut explosion — destruction was effectively saturated within the blast radius, leaving little additional upside risk in the tail. The

occurrence exceedance probability (OEP) curve confirms this: portfolio losses rise steeply between the 1-year and 10-year return periods and then plateau.

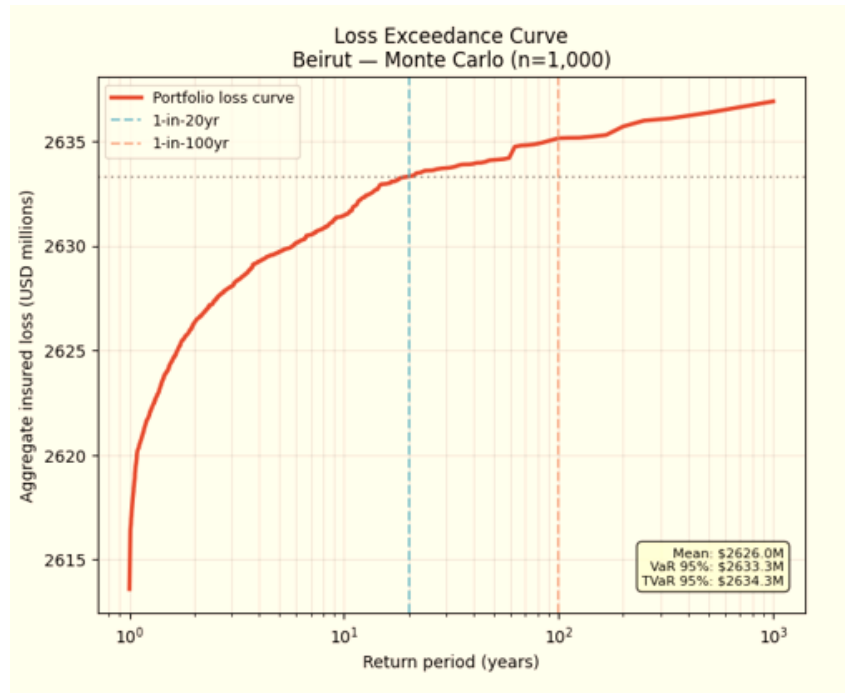


Figure 16: Loss Exceedance Curve

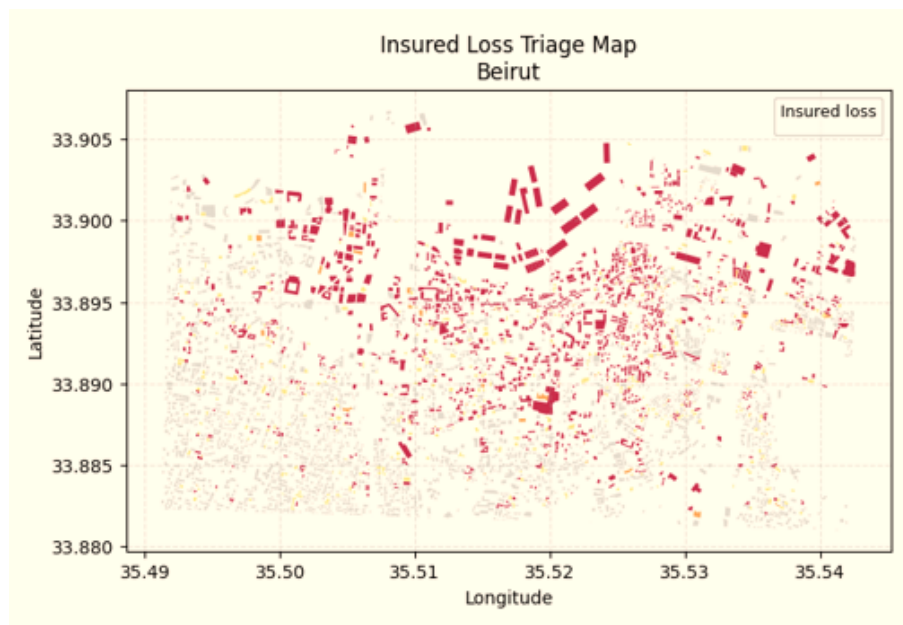


Figure 17: Triage Map

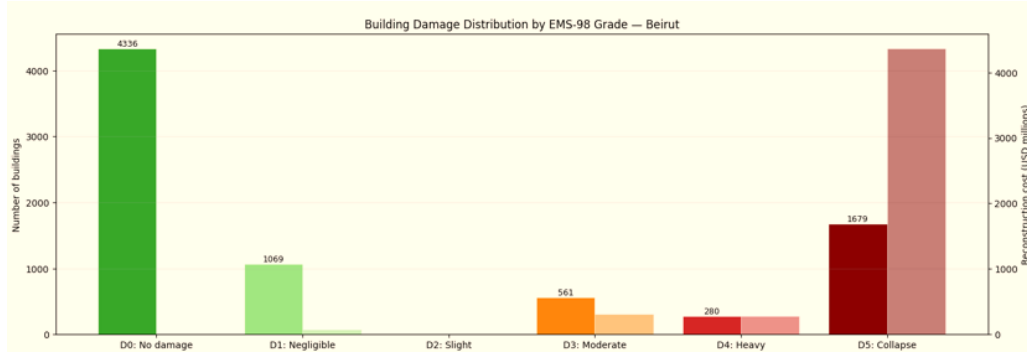


Figure 18: Damage grade distribution

Table 6: Beirut business intelligence summary: EMS-98 grade distribution, reconstruction cost, and insured loss. Grade counts are from all 7,293 buildings with valid HMP scores. Insurance figures apply to the 4,815 insured buildings (60% penetration). Reconstruction unit costs: \$1,100/m² (D5), \$750/m² (D4), \$400/m² (D3), \$50/m² (D1) per World Bank RDNA 2020 World Bank [2020]. All cost figures are P50 point estimates.

Grade	Label	Buildings	Share (%)	Notes
D5	Collapse	1,679	23.0	$p_B < 0.005$
D4	Heavy	280	3.8	$0.005 \leq p_B < 0.010$
D3	Moderate	561	7.7	$0.010 \leq p_B < 0.030$
D2	Slight	0	0.0	$0.030 \leq p_B < \tau^*$ (none)
D1	Negligible	1,069	14.7	$\tau^* \leq p_B < 0.10$
D0	Undamaged	3,704	50.8	$p_B \geq 0.10$
Total scored		7,293	100	
<i>Portfolio-level results (4,815 insured buildings at 60% penetration):</i>				
Reconstruction cost (P50)				\$5,022.6M
Reconstruction cost P10–P90 envelope				\$4,815.2M – \$5,295.3M
Claims triggered				1,517
Expected insured loss (mean)				\$2,626.0M
VaR _{0.95}				\$2,633.3M
VaR _{0.99}				\$2,635.0M
TVaR _{0.95}				\$2,634.3M

5. Commercial Applications and Business Model

The three-layer output — detection, reconstruction cost, and insured loss — maps onto distinct commercial use cases forming a **Post-Disaster Property Risk Intelligence Platform**.

Insurance loss estimation is the highest-value near-term application. Proprietary CAT models from firms such as RMS, AIR Worldwide, and Verisk are unavailable for active conflict zones where field adjusters cannot safely operate and optical imagery may be

obscured for days or weeks. The SAR-based approach is inherently suited to precisely these conditions: an insurer with policies in Mariupol had no way to assess physical losses from optical imagery alone.

Reconstruction monitoring enables development banks to verify that reconstruction loans are deployed as intended. Since the same coherence technique that detects collapse can detect rebuilding — coherence recovers toward pre-event levels as a newly stabilised structure replaces rubble — a monthly monitoring subscription provides independent verification of physical reconstruction progress.

Real-estate risk scoring targets investors and PropTech platforms in conflict-adjacent markets. A risk-adjusted land value layer combining damage density, proximity to critical infrastructure, and destruction probability provides pre-investment due diligence unavailable from any other source.

6. Limitations and Future Work

Several limitations warrant acknowledgement. The polynomial approximation to the penalised smoothing spline introduces a small approximation error relative to the R implementation in Racek et al. [2025]. The polynomial trend cannot fully capture highly non-linear seasonal coherence patterns in some environments, which may increase false positives in vegetated peri-urban areas. For Mariupol, the spread of destruction across many time periods means no single difference index has a dominant p-value signal, making building-level aggregation less decisive than in the Beirut case; our AUPRC of 0.377 falls below the ≈ 0.61 reported in the reference paper, likely attributable to the polynomial vs. spline regression difference.

The absence of the D2 damage grade in the Beirut output indicates that the $[0.03, \tau^*)$ range contains no buildings at the calibrated threshold, a consequence of the bimodal HMP distribution (damaged buildings cluster near zero, undamaged near 0.10–1.0). Future work should consider adaptive grade boundaries calibrated per event. The insurance portfolio is entirely synthetic; the reconstruction cost estimates are sensitive to the unit cost priors adopted from World Bank and UN-Habitat reports.

Future directions include: (i) acquisition of real insurer portfolio data under NDA; (ii) integration of Sentinel-1 backscatter intensity alongside coherence; (iii) real-time deployment via the Copernicus Emergency Management Service API; (iv) sensitivity analysis across varying seasonal conditions and building densities; and (v) applying the spline-based implementation (R/`quantreg`) to close the Mariupol performance gap.

7. Conclusion

This project demonstrates that an unsupervised SAR coherence analysis pipeline can reliably detect building destruction across three qualitatively distinct scenarios — a single-day point-source explosion (Beirut), a weeks-long siege (Mariupol), and a large-scale conflict (Gaza) — using only freely available Sentinel-1 imagery and OpenStreetMap building footprints. At the building level, detection achieves PR-AUC of 0.834 for Beirut and 0.920 for Gaza (with zero-shot threshold transfer), with the Gaza pre-war false-positive rate matching the reference paper at 0.021%. Mariupol’s building AUPRC of 0.377 reflects the methodological challenge of temporally diffuse conflict rather than pipeline failure.

The extended business intelligence module translates these statistical outputs into reconstruction cost estimates (Beirut P50: \$5.0B) and insurance loss distributions (Beirut expected loss: \$2.6B from 1,517 claims across 4,815 insured buildings), directly useful to commercial and humanitarian stakeholders. The narrow VaR–TVaR spread ($< \$2\text{M}$) demonstrates the actuarial concentration risk associated with a single-day catastrophic event.

The work underscores both the methodological maturity of SAR coherence for conflict damage assessment and the significant gaps in operationalising such tools at scale: access to real insurer portfolios, deeper calibration of damage grade mappings, and automated deployment pipelines remain important areas for future development.

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